

VEDA ACADEMIC ASSESSMENT SHEET

SUBJECT: SOFTWARE DEVELOPMENT

MAX MARKS: 20 PTS

CLASS/GRADE: JUNIOR WEB DEVS

DATE: 28/5/2026

STUDENT NAME: _____

ROLL NO / ID: _____

SECTION A: OBJECTIVE ASSESSMENT

Instruction: Choose the correct option below. Each question carries equal designated points.

- Q1.** Which functional interface introduced in Java 8 accepts a single argument, performs an action, and returns no result? **(2 Pts)**
- A) Function<T, R>
B) Supplier<T>
C) Consumer<T>
D) Predicate<T>
- Q2.** What is the primary difference between the map() and flatMap() methods in the Java Stream API? **(2 Pts)**
- A) map() is used for intermediate operations, while flatMap() is a terminal operation.
B) map() transforms each element of a stream into another element, while flatMap() transforms each element into a stream and flattens the resulting streams into a single stream.
C) map() executes synchronously, whereas flatMap() executes asynchronously.
D) map() maintains the original order of elements, whereas flatMap() does not.
- Q3.** In Java multi-threading, what is the significance of the 'volatile' keyword when applied to a member variable? **(2 Pts)**
- A) It guarantees that the variable is thread-safe and acts as a lock on the object.
B) It ensures that writes to the variable are immediately visible to all other threads and prevents instruction reordering on that variable.
C) It prevents the variable from being serialized during object transmission.
D) It makes the variable immutable after its first initialization.
- Q4.** Which JDBC interface should be used to execute parameterized SQL queries dynamically to prevent SQL injection vulnerabilities? **(2 Pts)**
- A) Statement
B) PreparedStatement
C) CallableStatement
D) ResultSet
- Q5.** In a Spring Boot application using Spring Data JPA, which annotation is typically used on a child entity class to establish a foreign key mapping back to a parent entity in a Many-To-One relationship? **(2 Pts)**
- A) @OneToMany
B) @JoinColumn
C) @MappedSuperclass
D) @PrimaryKeyJoinColumn
- Q6.** What does the @RestController annotation do in a Spring Boot application? **(2 Pts)**
- A) It configures the class to serve static HTML files using a view resolver.
B) It is a convenience annotation that combines @Controller and @ResponseBody, ensuring handler methods serialize return values directly into the HTTP response body.
C) It automatically registers the class as a database access repository bean.
D) It secures the endpoints within the class by applying default OAuth2 authentication.

SECTION B: SHORT-ANSWER QUESTIONS

Instruction: Write concise conceptual answers. Support statements with logical reasoning.

Q1. Explain the purpose of the `java.util.Optional` class introduced in Java 8. How does it help developers build safer web applications and prevent `NullPointerExceptions`? **(5 Pts)**

Q2. What is the difference between a 'fail-fast' iterator and a 'fail-safe' iterator in the Java Collections Framework? Provide one collection class example for each. **(5 Pts)**

Q3. Briefly describe the three primary states of an entity lifecycle within a JPA/Hibernate persistence context (Transient, Persistent, and Detached). **(6 Pts)**

Q4. What is the 'N+1 Select Problem' in JPA/Hibernate? Describe one strategy (e.g., using JPQL or EntityGraphs) to resolve it. **(5 Pts)**

Q5. Explain the role of a Filter (such as `OncePerRequestFilter`) in a Spring Web application. How does it differ from an Interceptor (`HandlerInterceptor`) in the request lifecycle? **(6 Pts)**

Q6. Explain how Spring Boot handles Dependency Injection (DI) under the hood using the `ApplicationContext`, and differentiate between constructor injection and field injection (`@Autowired` on fields). Why is constructor injection generally preferred in modern software development? **(6 Pts)**

SECTION C: DISCIPLINE PRACTICALS & CODING

Instruction: Provide clean and structured answers or complete executable codes.

- Q1.

Write a Java method using the Stream API that takes a List of integers, filters out the odd numbers, squares the remaining even numbers, and returns their sum. If the list is empty or contains no even numbers, return 0.

(10 Pts)

Method signature:
public int sumOfSquaresOfEvens(List<Integer> numbers)

- Q2.

Write a complete Java class representing a Spring Boot REST Controller. The class should map to '/api/greet', expose a single GET endpoint, accept an optional query parameter named 'name' (defaulting to 'World' if not provided), and return a JSON payload in the format: { "message": "Hello, [name]!" }.

(15 Pts)

You may use a Map<String, String> or a helper record/class to construct the JSON payload.

- Q3.

Write a Java method using JDBC that connects to a database, queries a table named 'employees' for columns 'id' (int) and 'name' (String) where 'salary' is strictly greater than a given threshold, and prints the results to the standard console. Ensure all resources (Connection, PreparedStatement, ResultSet) are closed safely using try-with-resources. Do not hardcode credentials.

(15 Pts)

Method signature:
public void printHighEarners(Connection conn, double minSalary) throws SQLException
